Jayant Jha

Data Engineer | Python | SQL | ETL/ELT | Big Data | Cloud Data Platforms

+91 9065927961 ◇ jayant20work@gmail.com ◇ Bengaluru, India ◇ LinkedIn

SUMMARY

Data Engineer with **3+ years of experience** designing, building, and optimizing **scalable ETL/ELT data pipelines** on **AWS and Azure cloud platforms**. Hands-on expertise in **Python, PySpark, SQL**, and **data warehousing**, leveraging **Data Engineering tools** to transform **raw data into analytics-ready datasets**. Experienced in **workflow orchestration** and **pipeline monitoring**, improving **data reliability, consistency, and performance**. Skilled in **data modeling**, **data ingestion**, and **cloud data platform optimization**, with a proven ability to support **business intelligence and analytics workloads** in **Agile, cross-functional environments**.

SKILLS

Programming & Scripting	Python, PySpark, SQL, C++
Data Engineering & Modeling	ETL / ELT Pipelines, Data Warehousing
AI Engineering	LLM Integration, RAG, Vector Embeddings
Data Processing & Databases	MySQL, Apache Spark
Workflow Orchestration	Apache Airflow, AWS Step Functions, SLA Management
Query Optimization	SQL Query Tuning, Indexing, Query Execution Plans, Performance Optimization
Tools & CI/CD	Git/GitHub, Bitbucket

EXPERIENCE

Data Engineer — Dec ’23 — Present
Genpact

- Designed scalable **ETL/ELT data pipelines using AWS, Azure, Databricks Lakehouse with Python**, implementing **Medallion Architecture (Bronze, Silver, Gold)** to transform raw ERP, procurement, and transactional data into analytics-ready datasets for **GE Vernova** renewable energy operations.
- Built ingestion and transformation workflows for multiple enterprise source systems including **SAP, ERP, DOOSAN**, processing procurement lifecycle data such as **Purchase Orders (POs), PO lines, shipments, receipts, Accounts Payable (AP), and distribution data** across large-scale supply chain workflows.
- Built **real-time and near real-time ingestion pipelines** using **AWS DMS (CDC)** from **MySQL RDS** into cloud data lakes, supporting scalable analytical and BI workloads.
- Developed **Python and PySpark**-based transformation workflows implementing **incremental loads, deduplication, SCD Type 2 logic, data harmonization, and business rule validations** across Bronze-to-Silver and Silver-to-Gold layers for downstream analytics and reporting.
- Automated and orchestrated enterprise workflows using **Apache Airflow DAGs**, YAML-based workflow configurations, CRON scheduling, retry handling, and Databricks job triggers, enabling scalable workflow scheduling, monitoring, and SLA management.
- Defined and revamped monitoring dashboards for **data pipelines in Apache Airflow, AWS Step Functions, Tableau**, improving data lifecycle visibility, operational monitoring, data quality scores by **15%**, and reducing data-related incidents.
- Redesigned and optimized **complex SQL transformation and outgestion workflows** using **DBeaver SQL jobs**, window functions, joins, unions, and subqueries, reducing query latency by **50%** and significantly improving the performance of critical enterprise data pipelines.
- Improved pipeline efficiency by migrating heavy filtering and validation conditions (including **cancel flags, imported flags, authorization status, and distribution-level logic**) upstream, reducing downstream processing overhead and contributing to major annual manual effort savings.
- Monitored production data pipelines and application health using Grafana dashboards, proactively identifying failures, performance bottlenecks, and SLA breaches to ensure high system availability.
- Collaborated with **cross-functional stakeholders** including data architects, analytics teams, visualization engineers, migration teams, and data scientists in Agile sprint cycles to deliver **end-to-end enterprise data solutions** supporting BI dashboards, reporting, and machine learning workloads.

SDE Intern-Data Engineering Associate — Jan ’23 — Dec ’23
Amadeus Labs

- Pioneered **backend data processing** components utilizing **Python, SQL, Bash, and PowerShell scripting**, supporting finance analytics and reporting systems, which increased data pipeline throughput by **20%** and improved query performance.
- Designed and **optimized SQL queries** for efficient **data retrieval and transformation** within relational databases, resulting in a **20% reduction in database load reducing** operational costs related to data storage.
- Centralized data pipeline code review practices, implementing **CI/CD and DevOps** automation, leading to a quantifiable **20% decrease in code defects** and elevating team velocity by **15% through streamlined workflows.**
- Containerized backend applications using Docker and deployed services on Kubernetes, improving deployment consistency, scalability, and development workflows.

PROJECTS

Conversational RAG System

- **Semantic Retrieval Pipeline**: Developed a **Conversational RAG** application using **Python, LangChain, ChromaDB,** and **Groq LLaMA**, enabling semantic retrieval from **PDF documents** and **YouTube transcripts**.
- **LLM Integration**: Built an end-to-end retrieval pipeline with **vector embeddings**, **context-aware prompt engineering**, and **LLM inference** to deliver grounded responses while minimizing hallucinations and improving answer relevance.

LLM Hallucination Detection Framework

- **Grounded LLM Evaluation**: Built an evaluation framework using **Python, LangChain, ChromaDB,** and **LLM evaluation metrics** to identify hallucinated responses by comparing generated outputs with retrieved contextual evidence.
- **Semantic Validation Pipeline**: Developed a **retrieval-based validation pipeline** leveraging **vector embeddings**, **semantic search**, and **grounded response evaluation** to improve factual consistency, trustworthiness, and explainability of LLM-generated answers.

Abalone Age Regressor (Machine Learning)

- Model Development & Optimization: Implemented regression with hyper-parameter tuning, evaluating performance across 3+ algorithms to improve prediction accuracy.
- Data Preprocessing & Feature Engineering: Performed data preprocessing and feature engineering, including normalization and outlier handling, improving model stability and performance on structured datasets with 4,000+ records.

EDUCATION

B.Tech, VIT Vellore (GPA: 8.2/10.0) — Jul ’19 — May ’23
Vellore, India

ACHIEVEMENTS / SCORES

LeetCode: Solved 500+ problems (Rating: 1615)
GeeksForGeeks: Ranked in Top 500 (Solved 200+ problems)